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# Recovery of Useful Energy from Lost Human Power in Gymnasium

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**Abstract**—Human powered gyms are ones in which equipment can be used to generate electricity. Since the concept of a green or sustainable gym is recent, the potential of cardio equipment like treadmills, stationary bikes and rowing machines in generating energy are discussed in this paper. The power generated in a typical human powered gym, based on the equipment already existing in the market is analyzed. The amount of carbon dioxide emissions saved and the payback period for implementing this gym are also estimated. A survey was carried out to better understand how people would react to the presence of such gym. It was found that the energy-producing equipment in the gym not only benefit environmentally but also economically in the long run.

## I. INTRODUCTION

Gyms or fitness centers are amongst the most inefficient places in terms of energy consumption. Not only do some exercising equipment like treadmills consume electricity, but the requirements for proper lighting and temperature which must serve to satisfy the comfort of the users also require a huge amount of energy. In many gyms around the world, entertainment options like T.V. shows or documentaries provided to the users are played on screen most of the time, thereby consuming large amount of energy. More commonly, users' own electronic equipment (whether mobile phones, mp3 players, iPods, etc) are also used while exercising, which consume some energy from the battery. Above all, most of the energy spent by the people working out are lost as useless heat, when these could actually be converted into useful energy.

The gym or "wellness center" located in Masdar Institute of Science and Technology, Abu Dhabi, United Arab Emirates is selected as the baseline for our study. Since the campus is located in a "low carbon" Masdar City, it is an excellent idea to use its gym for our study purpose. The scope relevant to this analysis, however, is only the energy consumption (especially from lighting and fan coil units) and energy production (from human-powered equipment) in the gym.

## II. LITERATURE REVIEW

### A. Green Gym

The requirements for making an existing gym turn green include using energy-producing equipment and consuming less electricity, which can be achieved by installation of better lighting systems and temperature controlling by proper use of air conditioning systems, consuming less water, etc. Some

examples of green gyms around the world are discussed in the case studies in section I-D.

### B. Gym Cardio Machines

1) *Treadmill*: A treadmill is not a self-powered machine, therefore it needs to be plugged in all the time even if no one is using it. This suggests that treadmills account for a large consumption compared to other gym equipment. According to [1], the treadmill was the third in terms of the overall energy consumption. For example, the power consumption for a treadmill with a speed of 5 mph is 480W and by increasing the speed to 9W increases the consumption to 1083W [2]. However, during standby, it consumes around 3 Watts.

A treadmill can be used to generate up to 100W of electricity per hour [3]. This number can vary according to the speed choice, resistance choice and the time. The higher any of these elements, the larger the power that can be harnessed.

Human powered treadmill is a device that converts human kinetic energy into electric potential energy. The energy exerted to turn a belt on the treadmill is transferred via a generator into a simple circuit that charges a small battery. The battery can be used to charge small electrical devices like laptops, mobiles etc. In order to build a human powered treadmill, a manual treadmill is used. The manual treadmill has two flywheels as a part of its front axle. Because it is a manual machine, no external power source is required to operate it. All power going into the system is from human power. The power coming out of the system is then collected in a 12V rechargeable battery. One flywheel of the treadmill's front axle is connected to the pulley, attached to the shaft of the dynamo, using belt drive system. Hence, the mechanical power produced by the treadmill's motion is converted to electrical energy with the help of DC electromagnetic dynamo.

2) *Elliptical machines*: Elliptical machines consist of machines like cross-trainers and stair steppers which simulate stair climbing and walking or running. These are self-powered and the energy that is not used to power the machine or the monitors can be harvested.

Depending on the speed and the resistance set by the user, the elliptical trainers offer a cardio-vascular workout ranging from minimal impacts to intensive ones. An hour exercise on average produce 100 Watts while an advanced exerciser can generate up to 150 Watts [4].

The kinetic energy from the workout can be converted to DC (direct current) when users use the elliptical machines. ReRev™ technology helps retrofit cardio equipment so that the DC can be converted to AC (alternating current). Each retrofit consists of a controller box which helps feed the current into a central-grid tied inverter that is connected to the building's electrical system [5]. A typical 30 minute workout will produce 50 Watt-hours of electric energy [6].

3) *Stationary bike*: A stationary bike is an exercise-equipment that consists of pedals, a saddle and handlebar, representing a bicycle but does not move from place to place. Some of the models have handlebars that are connected to the pedals so that besides the lower body, the upper body can also exercise at the same pace. The amount of energy generated from these devices varies according to user experience, user pace, and speed. Several estimates exist for the human generated power: 60-120W, 50W, 75-100W [7] and 150W [8].

One of the best technologies that can be done for energy harvesting by bikes has been developed by Latzke and Delcourt, called the Siva Cycle Atom [9]. Siva Atom is a device that can be mounted on a bicycle and its cost is USD 130. It converts momentum of the ride into power for electronics such as iPhone, Android Phones, lights, GPS and more. The capacity of the battery installed within the device is 1650MAH. Battery may be recharged from wall outlet with micro-USB in addition to directly from the Siva Atom. The battery itself can be charged from the ATOM or from any USB charging device. The lithium-polymer battery pack is rated for 2-4 years use. This quality is as high as the quality of batteries used in android and iPhone devices.

4) *Rowing machine*: A rowing machine simulates the body movement similar to that of rowing a boat, hence focusing on the upper parts of the body. 150 watts can be generated from 12 minutes exercise [10]. The coach at Williams College estimated that a man could create up to 200 watt hours while a woman could produce 160 watt hours [11].

In a rowing machine, the movement of the flywheel can be converted to electrical energy through the use of an alternator [10]. The alternator simply consists of a magnet that rotates as the rower perform his exercise. This part is called rotor. As the magnet rotates within a set of fixed conductors, called stator, current is generated. The faster the spin, the more is the electric current (AC) generated. In order for this energy to be useful, it will have to be converted to DC through a rectifier. The DC electrical power can either be used to charge batteries for different devices or again be converted to AC in order to power home appliances.

### C. Energy Floor

An average person takes around 150 million steps in a lifetime. The energy spent by a person stepping can be considered as energy lost which can be utilized so that it can generate electricity. This idea was implemented in several places including the Paris marathon in 2013 where 176

Pavegen tiles were installed across 25 meters generating 4.7 kilowatt-hours of energy which is enough to power five LED for 940 hours or to power a laptop for more than two days [12].

PaveGen [13] provides tiles that can be installed on the floor to harvest energy. These tiles if compressed to 5mm, convert the energy of the footsteps using the piezoelectric effect (deformation of the piezoelectric material result in charge accumulation in response to pressure) into electricity to be either directly fed into the grid or stored in a battery. The tiles are made from recycled polymer, with the top surface made from recycled truck tires. Each tile has a size of 450x600x68 mm and costs about USD 76 [14]. These tiles can be used in airport, schools, way finding lights, electronic chargers and street lights.

### D. Case Studies

1) *Recreational Sports Facility at University of California, Berkeley*: The basic purpose of this project was to analyze the technical and social benefits of human powered gym by installing 28 elliptical machines at Recreational Sports Facility (RSF) at University of California, Berkeley [15]. The plan to retrofit elliptical machines was based on the fact that human legs are four times powerful as compared to human arms. For the baseline case, the annual electricity data from 1986-2008 is taken. The average annual energy consumption was approximately 1.7 million kWh during these years. However, after certain improvements were made to the existing facility in the year 2008 and average annual energy consumption was calculated for the following year, there were energy savings of approximately 17.5% as compared to the baseline case i.e. the annual energy consumption was about 1.45 million kWh for the year 2008-2009. Following improvements were made to minimize annual energy consumption:

a) *Lighting retrofit*: The old lighting technology is replaced with the new efficient one and many areas were equipped with occupancy sensors at different locations to reduce the electricity consumption in unoccupied spaces. As a result of lighting retrofit, the power consumption by the lighting has been reduced approximately 25 percent, from about 730,000 kWh in 2007-2008, to about 580,000 kWh in 2008-2009.

b) *Energy harnessing*: The potential energy can be harnessed by retrofitting the existing 28 elliptical machines in RSF. The retrofitting can be made by replacing current built in resistance mechanism with a DC/AC micro-inverter. So on the basis of average human power potential of 100 W per day on an elliptical machine, the number of operational hours of RSF and system conversion efficiency of 85 percent, total energy production of 7,800 kWh was estimated. However, due to variability of usage in gym in different month of a year, it was estimated that 7,800 kWh is a minimum energy production (because it was recorded in July, when many students were away). Therefore, average energy consumption was estimated to be 29 percent more than this minimum value and was

around 10,000 kWh per year which accounts for the 1.1 percent of the energy consumed by the RSF.

2) *Williams College Fitness center*: The fitness center in the Williams College, referred to as “The Estrogym”, was one of the many buildings in college and was initially in a small building with few lights, very few machines and little need for high powered ventilation [4]. Due to limited space, Fitness center is shifted to a new room with greater capacity and hence greater energy requirements. Because of the creation of this new fitness center, College has bought new machines. Williams spent \$164,100 on twenty treadmills, \$112,900 on twenty Ellipticals, and \$25,560 on eight bikes. The elliptical machines and bikes are self-powered meaning that these machines use human power for their operation.

a) *Energy harnessing by elliptical machines*: To determine the weekly energy generation of machines, the following equation is used:

Energy generated = Number of Ellipticals x Watts x Hours weekly

Reasonable average of energy produced on an elliptical is 100 Watts.

Maximum potential expenditure on ellipticals:

$$20 \times 100 \times 91.5 = 183 \text{ kWh} \quad (1)$$

For half of the ellipticals utilized:

$$10 \times 100 \times 91.5 = 91.5 \text{ kWh} \quad (2)$$

However, the energy harnessed by this formula is based on certain assumptions and can be actually greater or smaller than this value based on resistance of the machine and time for exercise when the particular resistance is applied.

b) *Energy harnessing by stationary bikes*: For bikes, the same equation is used as in case of elliptical machines. i.e.

Energy generated = Number of Bikes x Watts x Hours weekly.

Reasonable average of energy produced on a bike is 100 Watts.

Maximum potential expenditure on bikes:

$$8 \times 50 \times 91.5 = 36.6 \text{ kWh} \quad (3)$$

For half of the bikes utilized:

$$4 \times 50 \times 91.5 = 18.3 \text{ kWh} \quad (4)$$

### III. METHODOLOGY

#### A. Survey

A survey is performed to collect data on gym usage from the student body. Questions regarding the frequency of gym users and their equipment of concern were addressed. By collecting this information, we will have an estimate of how much energy can be produced, based on their behavior of using the gym.

#### B. Determination of Power Consumption in the Gym

In order to calculate the electricity consumption in Gym, data is collected from Emrill (Contractors, Masdar Institute). The main sources of power consumption are lights, treadmills and fan coil units. The power rating of each equipment along with its number is shown in Table I.

In order to calculate the weekly power consumption of each equipment, certain procedure is developed with necessary assumptions.

1) *Lighting*: The type and number of lights are mentioned in Table I. The gym is basically divided into two areas for calculating power consumption by lighting: Active area and Changing room area. Active area comprises of Regular Gym, Functional Gym and reception areas. The division of these areas is based on the assumption that in all the active areas, the lighting usage will be the same and will be more than that in changing room area which is occasionally used.

All lights in the Gym area are controlled by occupancy sensors. The consumption will increase during the peak hours and will decrease during the off peak hours. In order to facilitate the calculation, the operating hours are divided in to two different parts, i.e. before 5 pm and after 5 pm.

Another assumption is made for the operation of light at one particular hour. It is assumed that at one particular hour, half of the lights are operational.

The total power consumption by different lights in the Gym using the above mentioned assumptions is shown in Table II.

2) *Fan coil units*: Fan coil units consume most of the power in the Gym. These are basically used to distribute the conditioned air to the space uniformly.

3) *Treadmills*: Out of all the powered exercising equipment in the Gym, Treadmills consume most power. Fig. 1 shows the percentage of weekly power consumption of the equipment mentioned above.

4) *Lighting retrofit*: Since there is sufficient sunlight in the Gym during day time so there is no need of lights during that time. The lights are required in the changing room area only during day time because it has no windows for sunlight. It is assumed that there is sufficient sunlight in active area before 5 p.m.

Table I. Power rating of different components

| Description         | Watts | No. Of equipment |
|---------------------|-------|------------------|
| Spot light          | 20    | 10               |
| Tube light 4 feet   | 28    | 56               |
| CFL lamps           | 18    | 82               |
| Tube light 2.5 feet | 21    | 10               |
| Tube light 5 feet   | 35    | 3                |
| FCU                 | 86    | 36               |
| Treadmill           | 1512  | 3                |

Table II. Weekly power consumption in the Gym

| Lights                | Weekly consumption in Watts |
|-----------------------|-----------------------------|
| Spot lights           | 5600                        |
| Tube light (4 feet)   | 43904                       |
| Tube light (2.5 feet) | 5880                        |
| Tube light (5 feet)   | 2940                        |
| CFL Lamps             | 19278                       |

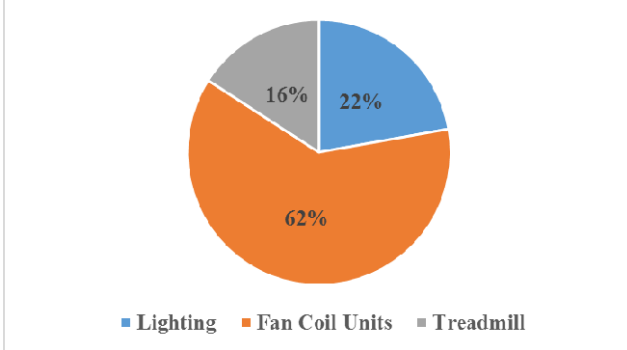


Fig. 1 Electricity consumption per week in the gym

Table III. Power produced by each machine

| Gym machines    | Power produced (Watts) |
|-----------------|------------------------|
| Treadmill       | 100                    |
| Stair-stepper   | 100                    |
| Cross-trainer   | 100                    |
| Stationary bike | 80                     |
| Rowing machine  | 68                     |

session at the gym are also collected, in order to calculate the amount of energy that could have been produce if the machines were human-powered ones. We also received positive responses regarding their increase and interest in use of the machines, if they were able to produce electricity. More than 80 % of the responses were willing to spend more time exercising in a human-powered gym.

The survey results also provided us the base for calculations of energy produced and amount of CO<sub>2</sub> emissions not-emitted. These can be found at “Energy generation” and “CO<sub>2</sub> emissions saved” in the sections below.

### C. Calculation of Energy Generation

1) *Cardio machines*: The amount of power that can be produced by each cardio machine is as tabulated in .

To calculate the amount of energy which can be generated from each of the machines, the following formula was used:

Energy generated (kWh/year) = Total number of hours using the machines per year × power generated by each machines.

2) *Energy floor*: Energy floor is one of the options discussed in this paper to generate electricity. We propose to cover an area of 605x200 cm<sup>2</sup> in the gym to harvest energy from the people working out or even footsteps energy. Sixteen tiles are required to cover that area with a cost of USD 1200. Assuming that 5 peak hours would result in 200 steps and 3 off-peak hours give 50 steps (where the amount of the energy produced per step is 10 watts), the following formula is used.

Energy generated (kWh/year) = yearly operational hours x footsteps per hour x power produced from one step

### D. Calculation of Carbon Dioxide Emissions Saved

Primary electricity production for UAE comes from natural gas. CO<sub>2</sub> emission factor for natural gas is 0.056 kg CO<sub>2</sub>/ MJ of NG [16] and 1kWh = 3.6 MJ. Therefore, the amount the gym can generate will be compared to that of the energy produced by natural gas, thereby observing how much CO<sub>2</sub> emissions can be saved.

## IV. RESULTS AND DISCUSSION

### A. Survey Results

The results collected form the survey gave us a rough estimate of the frequency of the gym usage for the student body of Masdar Institute of Science and Technology. Moreover, their usage of the different cardio-equipment per

### B. Energy Savings

After lighting retrofit, the weekly consumption of each light is reduced considerably. Fig. 2 and Fig. 3 below show the comparison of electricity consumption of each light with and without retrofit and percentage decrease in the lighting consumption after retrofit respectively.

### C. Energy Generation

1) *From cardio machines*: Fig. 4 shows the energy division generated from each machine in the gym. Treadmills generate 46% of the overall energy. After the treadmill comes the cross trainer. This can be attributed to the treadmill usage percentage in Masdar gym which constitute 42% of the overall weekly machine usage.

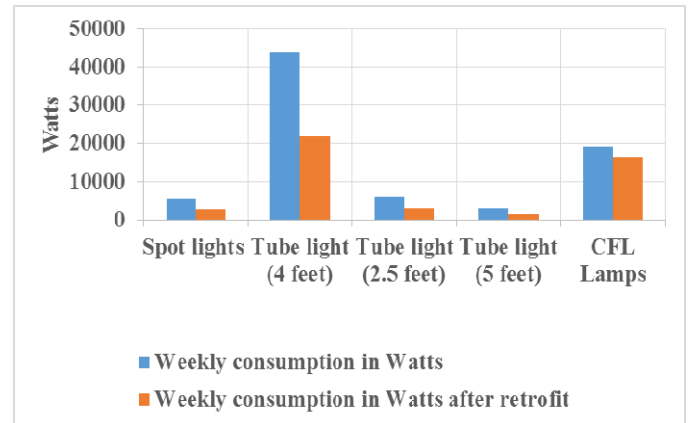


Fig. 2 Comparison before and after Lighting Retrofit

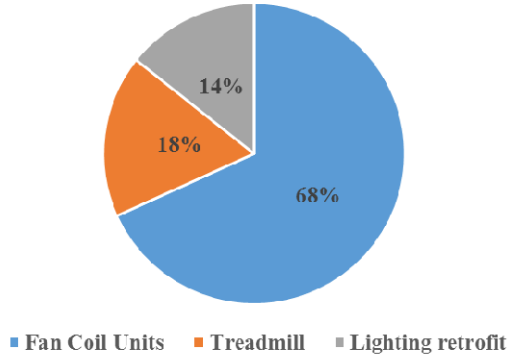


Fig. 3 Power consumption after Lighting Retrofit

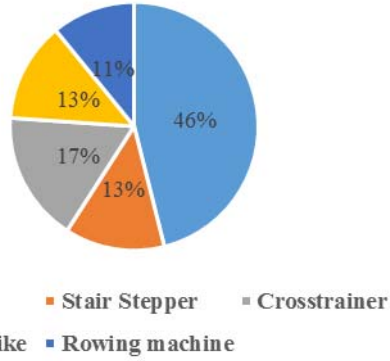


Fig. 4 Percent of energy produced yearly from each machine

The results from 53 people who answered our survey was used for the total energy and electricity which can be produced by each machine which is shown in Table IV. If we extrapolate the data and assume we have 150 people going to the gym, the total amount of energy generated can reach 1454.54 kWh in a year. For 300 gym participants the total estimated energy generated around 3000 KWh per year.

2) *From energy floor:* Table IV and Table V show the yearly power that can be generated from the gym machines and energy floors respectively and is based on our assumptions made in the methodology. Therefore, from the above results, the total energy that can be generated from the gym is 407.27 kWh + 4197.5 kWh = 4604.77 kWh/year.

#### D. Carbon Dioxide Emissions Saved

If an equal amount of natural gas is used to generate the energy equally to the ones from the human-powered gym machines results, i.e. 4604.77 kWh of NG produced 4604.77 kWh  $\times$  0.056 kgCO<sub>2</sub>/MJ  $\times$  3.6 MJ = 928.322 kg of CO<sub>2</sub>.

#### E. Payback Period

Based on the equipment in the gym in Masdar city and the technologies found in the literature, the estimates about the costs can be made as shown in Table VI.

Table IV. Energy produced per year

| Gym machines    | Energy produced per year in KWh |
|-----------------|---------------------------------|
| Treadmill       | 187.671                         |
| Stair-stepper   | 53.012                          |
| Cross-trainer   | 67.395                          |
| Stationary bike | 52.004                          |
| Rowing machine  | 32.654                          |
| Total           | = 407.272                       |

Table V. Yearly power generation from energy floors in KWh

|                | Operational hours | Footsteps per hour | Total energy generated weekly | Total energy generated yearly |
|----------------|-------------------|--------------------|-------------------------------|-------------------------------|
| Peak hours     | 5                 | 200                | 70                            | 3650                          |
| Off peak hours | 3                 | 50                 | 10.5                          | 547.5                         |
|                |                   |                    | Total                         | 4197.5                        |

Table VI. Approximate cost of equipment in USD

| Gym machines    | No. of equipment*Cost of equipment |
|-----------------|------------------------------------|
| Treadmill       | 3*5500                             |
| Stair-stepper   | 1200                               |
| Cross-trainer   | 2*1000                             |
| Stationary bike | 2*150                              |
| Rowing machine  | 7*150                              |
| Total           | = 21050                            |

The calculation for the payback period is based on the following assumptions:

- New treadmills were bought.
- “Siva Atom” technology is used for the bikes.
- Since no information is available about the cost of power generating rowing machine, the rowing machines also use Siva Atom’s technology
- The Cross-trainer can be purchased or upgraded to generate electricity with a cost of USD 1000.

According to DEWA (Dubai Electricity and Water Authority), the price of electricity in US dollars is 0.06\$/KWh [17].

Therefore, the amount generated from the gym, which is 4604.772 kWh/year can be converted to USD as:

$$4604.772 * 0.06 = \text{USD } 276 \text{ kWh/year.} \quad (5)$$

Thus, the payback period is the total cost of the equipment divided by the amount generated, i.e. 21050 / 276 = 75 years.

The payback period of 75 years calculated here may appear to be impractical. However, this can be attributed to two main reasons: the small sample size of 50 people in the survey conducted in this study and the high subsidy on electricity price from the government of Dubai. Since there are more people than only the 50 participant sample size in this survey, more energy can be generated and hence the payback period can also be lowered. For example, if the number of people using the gym is 150 people, the power generated is tripled, and the payback period reduces to 33.3% of its value, i.e. 25 years in UAE. Also, if the same study is applied to a

gym in a country like Italy where the electricity cost is 0.2839 USD/kWh, then the payback period could be reduced to 16 years for 50 people assumption and can go to as low as 5 years for 150 people.

#### CONCLUSION AND RECOMMENDATIONS

In this study we look into options of improving an energy-consuming gym to an energy-producing one, thereby promoting the sustainability of the recreation and health facilities. It was found that the energy-producing equipment in the gym not only benefit environmentally but also economically in the long run. From this study, the results show that with just a small number of people, the energy produced from the gym per year can be used to drive a NISSAN LEAF for over 13000 miles. Economically, it is also quite feasible since the payback period is not too long.

Since the calculation was based only on the responses we received from the survey, this amount of energy production can go higher with more number of responses. The amount of the energy generated from the given number of gym users were impressive and shows that the success of implementing such gyms is not a myth anymore. With the advancements in technology, all gym machines in the near future could be used as energy sources, which can benefit the environment, society and economy in the long run.

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